HUMAN ORGAN SYSTEMS AND BIO DESIGNS (QUALITATIVE)- PART A

## 3. Brain as a CPU system

The brain does not have a CPU. Instead, it has millions of neurons that combine signals simultaneously. At any given time, many large, specialized areas of the brain are operating in parallel to perform a variety of tasks, such as processing visual or auditory information or planning an action.

Computers are deterministic machines in the sense that with a given input, they will always produce the same output. This does not mean that this output is always predictable. For example, computers can simulate non-deterministic systems by introducing pseudo-random variables. Computers can also apply equations from chaos physics, in which the results of deterministic processes can be greatly influenced by tiny variations in the initial conditions.

The brain as a whole is considered a non-deterministic system, for the very simple reason that it is never completely the same from one moment to the next. It is constantly forming new synapses and strengthening or weakening existing ones according to how they are being used. Consequently, a given input will never produce exactly the same output twice. However, the physiochemical processes underlying brain activity are considered to be deterministic.

### 3.1 Architecture

Brains are built over time, from the bottom up. The basic architecture of the brain is constructed through an ongoing process that begins before birth and continues into adulthood. Simpler neural connections and skills form first, followed by more complex circuits and skills. In the first few years of life, more than 1 million new neural connections form every second. After this period of rapid proliferation, connections are reduced through a process called pruning, which allows brain circuits to become more efficient.

Brain architecture is comprised of billions of connections between individual neurons across different areas of the brain. These connections enable lightning-fast communication among neurons that specialize in different kinds of brain functions. The early years are the most active period for establishing neural connections, but new connections can form throughout life and unused connections continue to be pruned. Because this dynamic process never stops, it is impossible to determine what percentage of brain development occurs by a certain age. More importantly, the connections that form early provide either a strong or weak foundation for the connections that form later.

The interactions of genes and experience shape the developing brain. Although genes provide the blueprint for the formation of brain circuits, these circuits are reinforced by repeated use. A major ingredient in this developmental process is the serve and return interaction between children and their parents and other caregivers in the family or community as shown in Fig.3.1. In the absence of responsive caregiving—or if responses are unreliable or inappropriate—the brain's architecture does not form as expected, which can lead to disparities in learning and behaviour. Ultimately, genes and experiences work together to construct brain architecture.

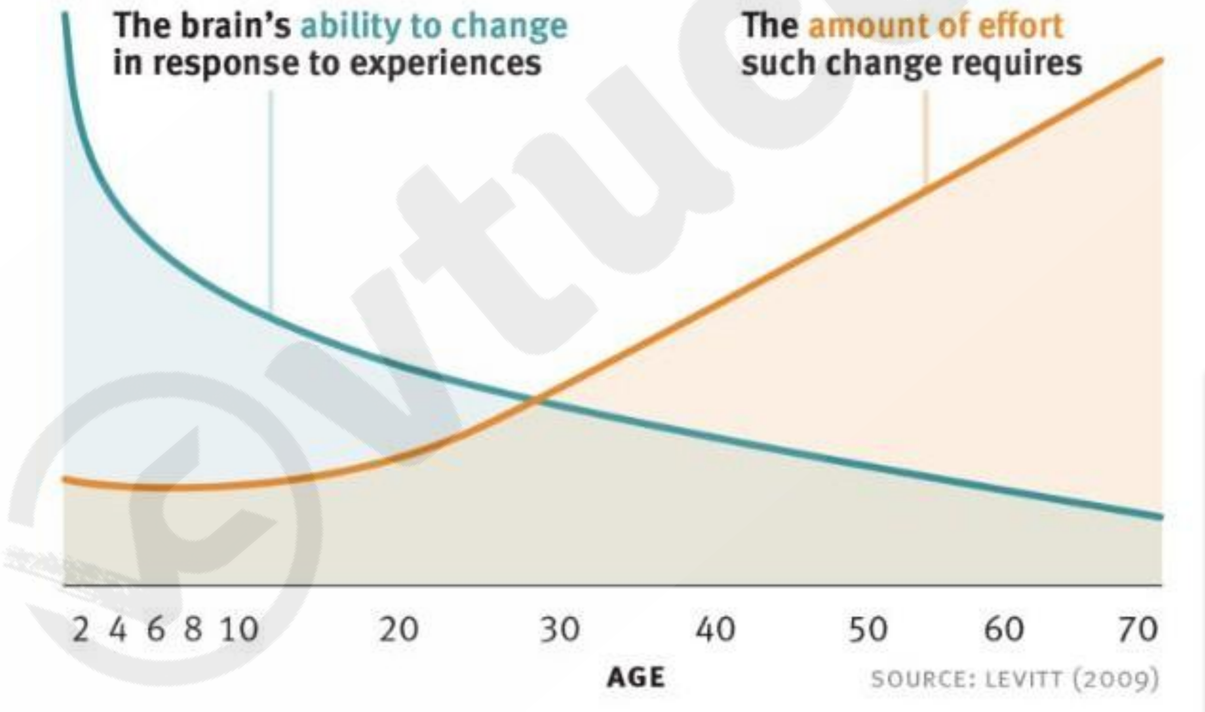


Fig.3.1 Brain development over the years

## 3.2 CNS and Peripheral Nervous System

### 3.2.1 CNS (central nervous system)

Our bodies couldn't operate without the nervous system - the complex network that coordinates our actions, reflexes, and sensations. Broadly speaking, the nervous system is organised into two main parts, the central nervous system (CNS) and the peripheral nervous system (PNS).

The CNS is the processing centre of the body and consists of the brain and the spinal cord. Both of these are protected by three layers of membranes known as meninges. For further protection, the brain is encased within the hard bones of the skull, while the spinal cord is protected with the bony vertebrae of our backbones. A third form of protection is cerebrospinal fluid, which provides a buffer that limits impact between the brain and skull or between spinal cord and vertebrae.

- **Grey and white matter**

In terms of tissue, the CNS is divided into grey matter and white matter. Grey matter comprises neuron cell bodies and their dendrites, glial cells, and capillaries. Because of the abundant blood supply of this tissue, it's actually more pink-coloured than grey.

In the brain, grey matter is mainly found in the outer layers, while in the spinal cord it forms the core 'butterfly' shape as shown in Fig3.2.

Fig.3.2 Grey and white matter

White matter refers to the areas of the CNS which host the majority of axons, the long cords that extend from neurons. Most axons are coated in myelin - a white, fatty insulating cover that helps nerve signals travel quickly and reliably. In the brain, white matter is buried under the grey surface, carrying signals across different parts of the brain. In the spinal cord, white matter is the external layer surrounding the grey core.

**3.3 The brain**

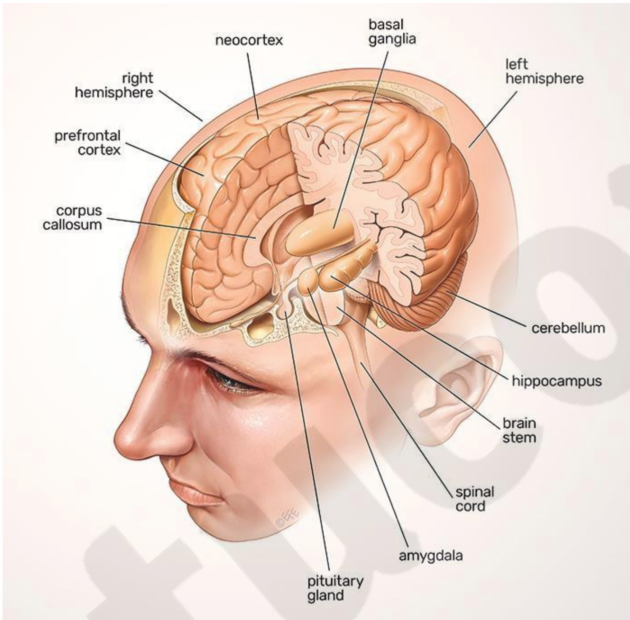


Fig.3.3 Brain structure

If the CNS is the processing center of the human body, the brain is its headquarters. It is broadly organized into three main regions - the forebrain, the midbrain, and the hindbrain. The largest of these three is the forebrain (derived from the prosencephalon in the developing brain). It contains the large outermost layer of the brain, the wrinkly cerebral cortex, and smaller structures towards its center, such as the thalamus, hypothalamus, and the pineal gland as shown in Fig.3.3.

The midbrain (derived from the mesencephalon in the developing brain) serves as the vital connection point between the forebrain and the hindbrain. It's the top part of the brainstem, which connects the brain to the spinal cord.

The hindbrain (derived from the rhombencephalon in the developing brain) is the lowest back portion of the brain, containing the rest of the brainstem made up of medulla oblongata and the pons, and also the cerebellum - a small ball of dense brain tissue nestled right against the back of the brainstem.

### 3.3.1 Peripheral Nervous System

Your peripheral nervous system (PNS) is that part of your nervous system that lies outside your brain and spinal cord as shown in Fig. 3.4. It plays key role in both sending information from different areas of your body back to your brain, as well as carrying out commands from your brain to various parts of your body.

Some of those signals, like the ones to your heart and gut, are automatic. Others, like the ones that control movement, are under your control.

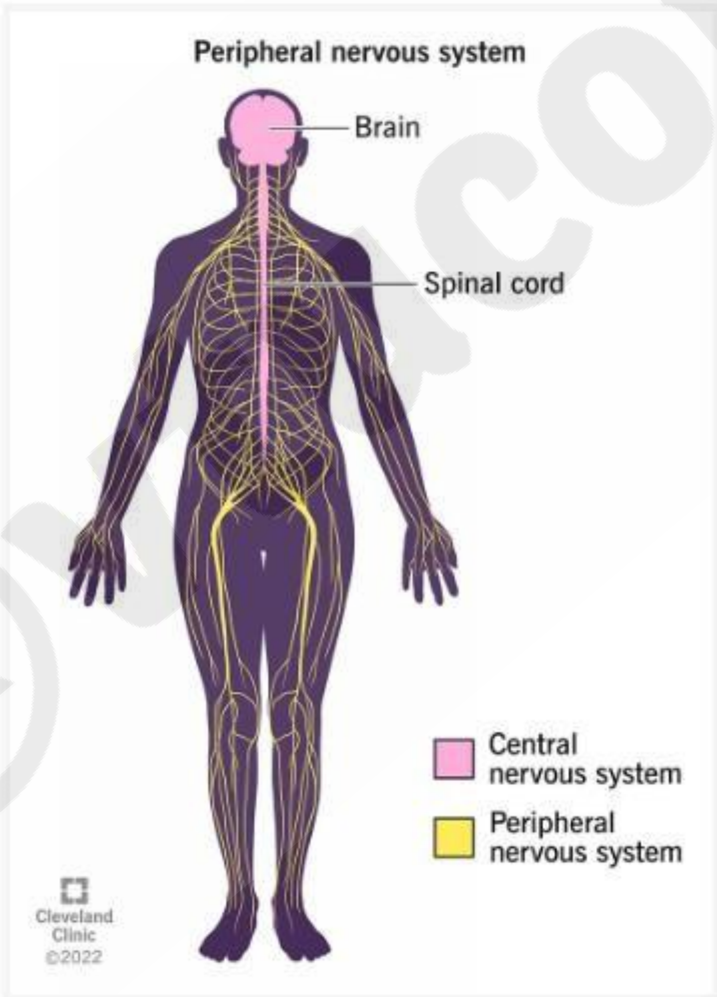


Fig.3.4 Peripheral Nervous System

Difference between the peripheral and central nervous systems

Your nervous system consists of two main parts: your central nervous system and your peripheral nervous system. Your central nervous system includes two organs, your brain and spinal cord.

Your peripheral nervous system is everything else and includes nerves that travel from your spinal cord and brain to supply your face and the rest of your body. The term “peripheral” is from the Greek word that means around or outside the center.

- **Signal transmission**

Information to and from the brain travels along neurons which are arranged in networks that let them pass information between the body and the brain. Here are the basics:

- Information is sent as packets of messages called action potentials.
- Action potentials travel down a single neuron cell as an electrochemical cascade, allowing a net inward flow of positively charged ions into the axon.
- Within a cell, action potentials are triggered at the cell body, travel down the axon, and end at the axon terminal.
- The axon terminal has vesicles filled with neurotransmitters ready to be released.
- The space between the axon terminal of one cell and the dendrites of the next is called the synapse.

### 3.3.2 Electroencephalogram (EEG)

An electroencephalogram (EEG) is a test that measures electrical activity in the brain using small, metal discs (electrodes) attached to the scalp as shown in Fig.3.5. Brain cells communicate via electrical impulses and are active all the time, even during asleep. This activity shows up as wavy lines on an EEG recording.

An EEG is one of the main diagnostic tests for epilepsy. An EEG can also play a role in diagnosing other brain disorders.

An EEG can find changes in brain activity that might be useful in diagnosing brain disorders, especially epilepsy or another seizure disorder. An EEG might also be helpful for diagnosing or treating:

- Brain tumors
- Brain damage from head injury
- Brain dysfunction that can have a variety of causes (encephalopathy)
- Sleep disorders
- Inflammation of the brain (herpes encephalitis)
- Stroke
- Sleep disorders
- Creutzfeldt-Jakob disease

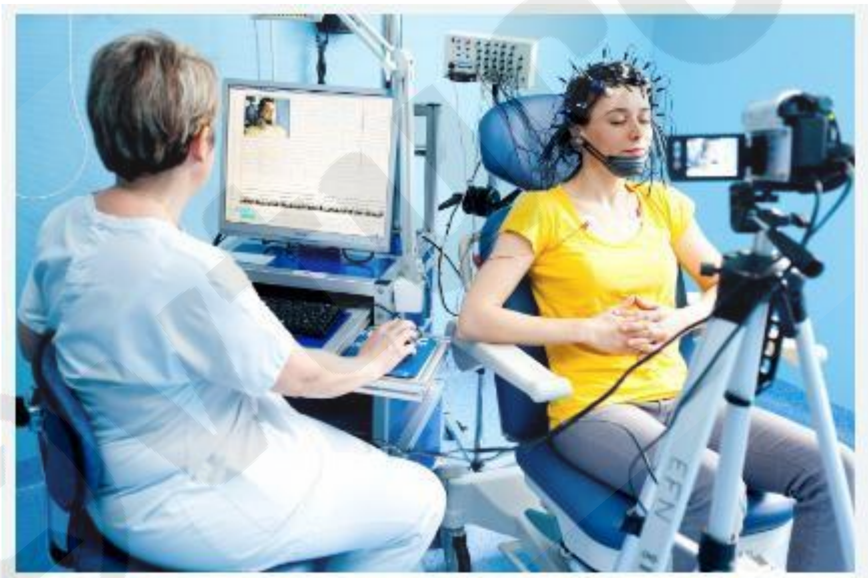

Fig.3.5 Electroencephalogram

**3.4 Robotic arms for prosthetics**

The idea of a robotic-powered limb may sound like something from a sci-fi film. The phrase conjures up images of super intelligent cyborgs, in which humans transcend the limitations of biology by merging with machines. While that scenario is some way off for now, the bionic limbs themselves are more than a pipe dream. Increasingly, prosthetics are being integrated with advanced robotics, with a view to restoring the functionality of a healthy human limb as shown in

Fig.3.6.

These devices are enabling patients to regain their motor capabilities and even restore sensory feedback – and many of the devices under development are mind-controlled.

“Only a few years ago, robotic prostheses could only do basic tasks, like standing up and walking at a constant speed indoors. Now they can perform many complex movements, similar to biological legs. They can walk on inclined or rough terrains, climb stairs, squat, lunge, and even run.”

The need for these kinds of devices is not in doubt. Millions of people around the world live with limb loss, often due to diabetic neuropathy or trauma. Around 185,000 amputations are performed annually in the US, along with 431,000 in Europe.

For these patients, their quality of life will depend to a large degree on the quality of their prosthetic devices. Traditional passive prostheses may give them some mobility, but they can’t restore natural muscle function and often end up restricting the person’s activity levels.

Fig.3.6 Prosthetics arms

### 3.4.1 Engineering solutions for Parkinson's disease

Parkinson's disease is a progressive disorder that affects the nervous system and the parts of the body controlled by the nerves. Symptoms start slowly. The first symptom may be a barely noticeable tremor in just one hand. Tremors are common, but the disorder may also cause stiffness or slowing of movement.

MRI-guided focused ultrasound (MRgFUS) is a minimally invasive treatment that has helped some people with Parkinson's disease manage tremors. Ultrasound is guided by an MRI to the area in the brain where the tremors start. The ultrasound waves are at a very high temperature and burn areas that are contributing to the tremors.

### 3.5 Eye as a Camera system

There are many similarities between the human eye and a camera, including a diaphragm to control the amount of light that gets through to the lens. This is the shutter in a camera, and the pupil, at the centre of the iris, in the human eye. A lens to focus the light and create an image. The image is real and inverted. A method of sensing the image. In a camera, the film is used to record the image; in the eye, the image is focused on the retina, and a system of rods and cones is the front end of an image-processing system that converts the image to electrical impulses and sends the information along the optic nerve to the brain. The way the eye focuses light is interesting because most of the refraction that takes place is not done by the lens itself, but by the aqueous humour, a liquid on top of the lens. Light is refracted when it comes into the eye by this liquid, refracted a little more by the lens, and then a bit more by the vitreous humour, the jelly-like substance that fills the space between the lens and the retina.

### 3.5.1 Architecture of rod and cone cells

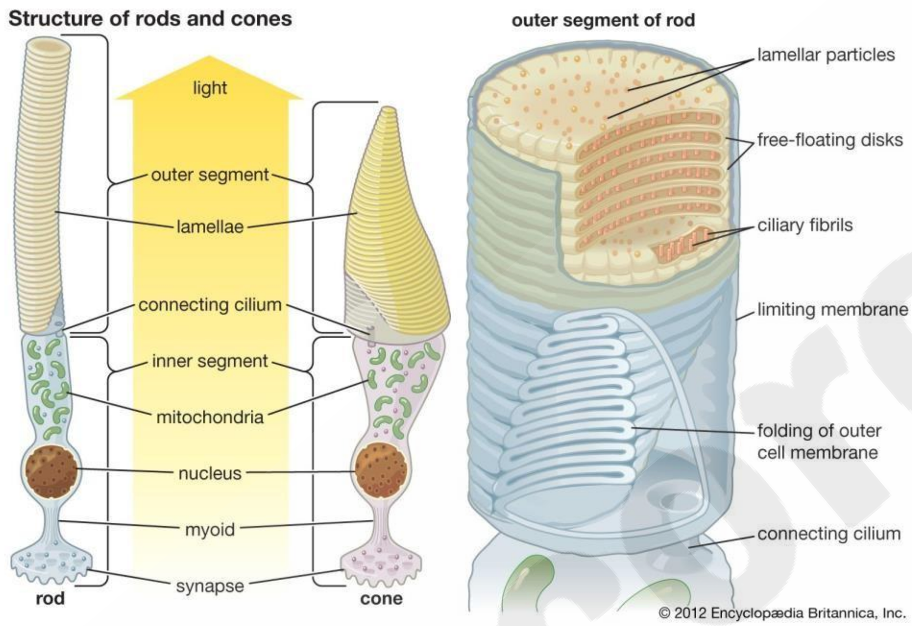


**Fig.3.7 Prosthetics arms**

Photoreceptors are the cells in the retina that respond to light. Their distinguishing feature is the presence of large amounts of tightly packed membrane that contains the photopigment rhodopsin or a related molecule. The tight packing is needed to achieve a high photopigment density, which allows a large proportion of the light photons that reach the photoreceptor to be absorbed. Photon absorption contributes to the photoreceptor's output signal.

In the retina of vertebrates, the rods and cones have photopigment-bearing regions (outer segments) composed of a large number of pancakelike disks. In rods the disks are closed, but in cones the disks are partially open to the surrounding fluid. In a typical rod there are about a thousand disks, and each disk holds about 150,000 rhodopsin molecules, giving a total of 150 million molecules per rod as shown in Fig.3.7. In most invertebrate photoreceptors the structure is different, with the photopigment borne on regularly arranged microvilli, fingerlike projections with a diameter of about 0.1 µm. This photoreceptor structure is known as a rhabdom. The photopigment packing is less dense in rhabdoms than in vertebrate disks. In both vertebrate photoreceptors and rhabdoms, each photoreceptor cell contains a nucleus, an energy-producing region with mitochondria (in the inner segment in rods and cones), and an axon that conveys electrical signals to the next neurons in the processing chain. In reptiles and birds the receptors may also contain coloured oil droplets that modify the spectrum of the light absorbed by the photopigment, thereby enhancing colour vision. In insects and other invertebrates the receptors may also contain granules of dark pigment that move toward the rhabdom in response to light. They act as a type of pupil, protecting the rhabdom in bright conditions by absorbing light.

### 3.5.1 Optical corrections

When your vision is blurry or fuzzy, it can make it difficult to go about your daily activities. Fortunately, though, there are effective ways of correcting your vision and seeing clearly. Many times, blurry or unclear vision is caused by what's known as a refractive error. A refractive error occurs when light doesn't bend correctly as it enters your eye. If the light that enters your eye doesn't hit your retina correctly — the light-sensitive tissue at the back of your eye — your vision won't be sharp. No matter what type of refractive error you have, vision correction can help you see clearly. Finding the right vision correction option allows you to drive, read, work on a computer, and do other daily tasks without squinting or straining your eyes to see properly. For many people with refractive errors, these tasks would be impossible without some type of vision correction. Below are four such processes that are recommended by the doctors in India for Optical corrections.

### 3.5.2 Refractive Lens Exchange

This procedure is recommended to correct the spectacle power in people over the age of 40 years. After the age of 40 years, the natural lens of the eye becomes hard and non-flexible and therefore loses its ability to change focus from distance to reading.

An eye is like a camera with a lens focusing light on the retina. Glasses are required if the image is out of focus. In RLE the natural lens is replaced by an artificial lens of suitable power to bring the image to focus on the retina without the need of glasses.

### 3.5.3 Monovision

Presbyopia or decrease in near vision with age. In a young individual, the natural lens of the eye is soft and flexible and works like an auto-focus lens. By contraction of the internal eye muscles, it can change its power from a distance to near.

Around the age of 45, the natural lens of the eye undergoes age changes and tends to become hard and non-flexible. Even with contraction of internal eye muscles, it is unable to change its power from a distance to near therefore additional spectacle power is required for reading purposes.

In the elderly, the eye requires a distance and a near spectacle power. The laser can correct only one power either for distance or for near. In mono-vision, one eye is corrected fully to focus it for distance, and other, partially for near. Thus with both eyes open, the person can do most of his routine activities without glasses and only for very fine work does a person need glasses.

### 3.5.4 Corneal Inlays

Lots of research and innovations are in progress to overcome near vision handicap. One of the recent innovations is in the form of corneal inlays.

By placing inlays in the cornea, the near vision is enhanced, without losing the advantage of distance vision without glasses.

### 3.5.5 Intacs

Intacs is the face of modern technology for Refractive Correction. It is useful in low myopia -1.0D to -3.0D an, however, in low myopia the choice of refractive correction is LASIK.

Intacs have proven value by centering the cone, reducing the cylinder power and improving the quality of vision (by reducing aberration at the level of the cornea).

### 3.5.6 Cataract

A cataract is a dense, cloudy area that forms in the lens of the eye. A cataract begins when proteins in the eye form clumps that prevent the lens from sending clear images to the retina. The retina works by converting the light that comes through the lens into signals. It sends the signals to the optic nerve, which carries them to the brain.

It develops slowly and eventually interferes with your vision. You might end up with cataracts in both eyes, but they usually don’t form at the same time. Cataracts are common in older people.

Common symptoms of cataracts include:

- blurry vision
- trouble seeing at night
- seeing colors as faded
- increased sensitivity to glare
- halos surrounding lights
- double vision in the affected eye
- a need for frequent changes in prescription glasses

**3.5.7 Causes of Cataracts**

There are several underlying causes of cataracts. These include:

- an overproduction of oxidants, which are oxygen molecules that have been chemically altered due to normal daily life
- smoking
- ultraviolet radiation
- the long-term use of steroids and other medications
- certain diseases, such as diabetes
- trauma
- radiation therapy

**3.5.8 Risk Factors of Cataracts**

Risk factors associated with cataracts include:

- older age
- heavy alcohol use
- smoking
- obesity
- high blood pressure

- previous eye injuries
- a family history of cataracts
- too much sun exposure
- diabetes
- exposure to radiation from X-rays and cancer treatments

### 3.5.9 Lens materials

- **Standard glass**

Glass had been the material most widely used for ocular lenses until the 1970s. Glass provides superior optical quality and has the most scratch-resistance surface, however it has several limitations including heavy weight, increased thickness, and low impact resistance. Glass lenses must be treated to comply with the American National Standards Institute impact resistance standards.[17] Chemical or thermal tempering can increase the shatter resistance, however this effect is lost if the lens is scratched or worked on with any tool after tempering. Individuals with myopia who desire thin glasses may opt for high-index glass, however the highest-index glass lenses cannot be tempered and require patients to sign a waiver accepting the risk of breakage. Additionally, high-index glass does not block ultraviolet light without a coating.

- **Standard plastic**

Plastic lenses gained popularity in the 1970s and have the benefits of weighing half as much as glass lenses due to their lower specific gravity and having high optical quality.[17] CR-39 which stands for Columbia resin #39 is the most commonly used plastic polymer lens material. The lenses block 80% of ultraviolet light without treatment, and can be tinted and coated to provide further ultraviolet light blocking. Plastic lenses tend to have a lower index of refraction, which require thicker lenses. The lens surface is also softer and thus easier to scratch, however scratch-resistant coatings are available to create a harder surface. CR-39 lenses in particular do not have the shatter resistance of polycarbonate or Trivex, increasing risk to wearers.

- **Polycarbonate**

High index polycarbonate lenses were popularized in the 1980s due to light weight, thin profile, superior impact resistance, and ultraviolet protection. These lenses are often recommended for children, young adults, individuals with active lifestyles, and as safety eyewear. They are very durable and can be up to 30% thinner than regular glass or plastic lenses. Disadvantages include high chromatic aberration indicated by its low Abbe number, which results in color fringing most noticeable in strong prescriptions. Additionally, polycarbonate is the most easily scratched plastic, thus requiring a scratch-resistant coating.

- **Trivex**

Trivex was introduced in 2001 and is a highly impact-resistant material with a low specific gravity delivering strong optical quality and minimal chromatic aberration indicated by its high Abbe number. Trivex is also able to block nearly all ultraviolet light. A disadvantage of Trivex lenses is its low index of refraction, thus requiring thicker lenses for higher powers. At the ±3.00 Diopter prescription range, this material allows for a comparably thin lens. Trivex is the lightest material available and meets high-velocity impact standards. A scratch-resistant coating is required for this lens.[16]

- **High-index materials**

High-index materials are defined by a refractive index of 1.60 or higher, and can be either glass or plastic. The main utility of high-index lenses is for high-power prescriptions to create thin and cosmetically attractive lenses.[16] The weight, optical quality, and impact resistance of high-index lenses vary based on the material used. For high-index glass, the specific gravity tends to run high, which means that these lenses are often heavy compared to other materials. None of the high-index materials pass the American National Standards Institute’s impact resistance standards.[16]

- **Bionic eye**

A bionic eye, also called a visual prosthesis, is an electrical implant that is surgically inserted into the eye. It improves light sensitivity and creates a sense of vision for people who have advanced vision loss. Most of the devices being developed are for individuals who have retinal degeneration caused by diseases like retinitis pigmentosa (RP) and age-related macular degeneration (AMD). There are other devices being studied that bypass the optic nerve and may be useful for people who have other types of vision loss

**3.6 Heart as a pump system**

Heart is a single organ, but it acts as a double pump. The first pump carries oxygen-poor blood to your lungs, where it unloads carbon dioxide and picks up oxygen. It then delivers oxygen-rich blood back to your heart. The second pump delivers oxygen-rich blood to every part of your body.

3.6.1 Architecture of Heart

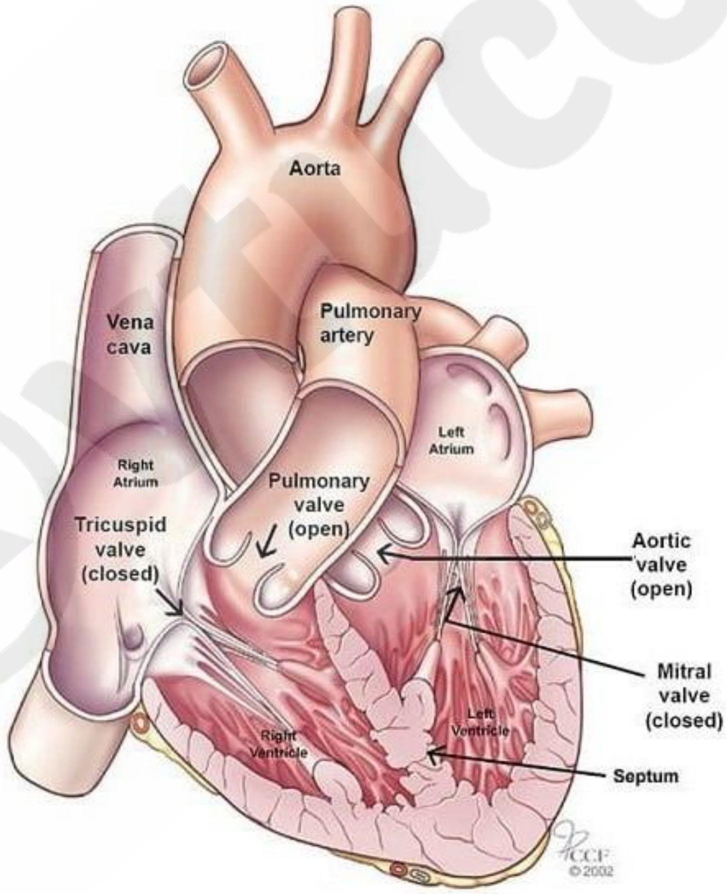


**Fig 3.8 Inside of the Heart**

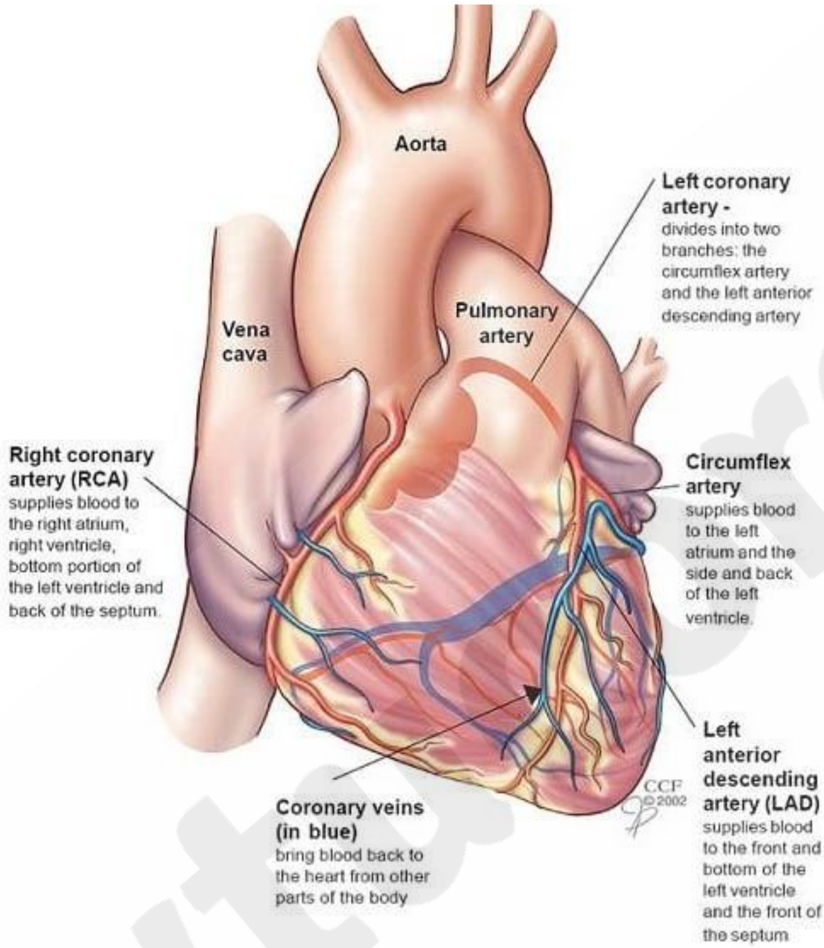


**Fig.3.9 Outside of the Heart**

Your heart is divided into four chambers as shown in Figs.3.8 & 3.9. You have two chambers on the top (atrium, plural atria) and two on the bottom (ventricles), one on each side of the heart.

- **Right atrium:** Two large veins deliver oxygen-poor blood to your right atrium. The superior vena cava carries blood from your upper body. The inferior vena cava brings blood from the lower body. Then the right atrium pumps the blood to your right ventricle.
- **Right ventricle:** The lower right chamber pumps the oxygen-poor blood to your lungs through the pulmonary artery. The lungs reload blood with oxygen.
- **Left atrium:** After the lungs fill blood with oxygen, the pulmonary veins carry the blood to the left atrium. This upper chamber pumps the blood to your left ventricle.
- **Left ventricle:** The left ventricle is slightly larger than the right. It pumps oxygen-rich blood to the rest of your body.

**Electrical signalling - ECG monitoring and heart related issues**

An electrocardiogram is a painless, noninvasive way to help diagnose many common heart problems. A health care provider might use an electrocardiogram to determine or detect:

- Irregular heart rhythms (arrhythmias)
- If blocked or narrowed arteries in the heart (coronary artery disease) are causing chest pain or a heart attack
- Whether you have had a previous heart attack
- How well certain heart disease treatments, such as a pacemaker, are working

**ECG recording**

- You'll have three electrodes put on your chest and the wires attached to these will be taped down.
- You'll wear a small portable recorder on a belt around your waist which the wires will lead to.
- While you're wearing the ECG recorder, you can do everything you would normally do - except have a bath or shower.
- It's safe and completely painless.
- When the test is finished, you'll return the recorder to the hospital so the results can be analyzed.

**Reasons for blockages of blood vessels**

Coronary artery disease starts when fats, cholesterols and other substances collect on the inner walls of the heart arteries as shown in **Fig.3.11**. This condition is called atherosclerosis. The buildup is called plaque. Plaque can cause the arteries to narrow, blocking blood flow. The plaque can also burst, leading to a blood clot.

Besides high cholesterol, damage to the coronary arteries may be caused by:

- Diabetes or insulin resistance

- High blood pressure
- Not getting enough exercise (sedentary lifestyle)
- Smoking or tobacco use

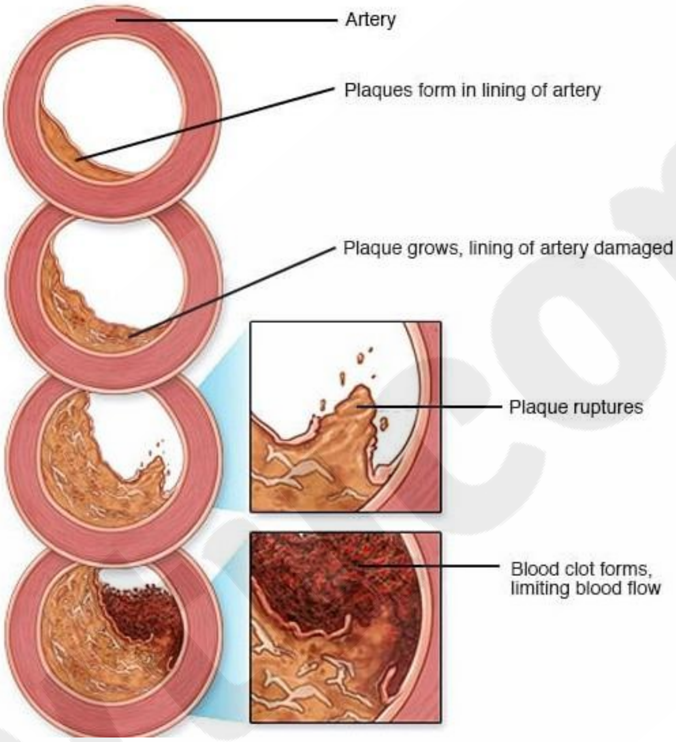


**Fig.3.11 Blockages of blood vessels**

**Design of stents**

Heart stents are placed inside the artery at the area of blockage or weakness to decrease the blockage or strengthen the artery wall. Stents can be placed primarily or after balloon angioplasty. Most often, they are made of metal mesh. There are two types of stents: bare-metal stent and drug-eluting stent. The latter are used more frequently and are coated with medication that helps keep a blocked artery open longer. The stent eventually becomes a part of the artery that it's placed in.

**Pace makers**

A pacemaker is a small device that's placed (implanted) in the chest to help control the heartbeat. It's used to prevent the heart from beating too slowly. Implanting a pacemaker in the chest requires a surgical procedure.

A pacemaker is also called a cardiac pacing device.

**Types**

Depending on your condition, you might have one of the following types of pacemakers.

- **Single chamber pacemaker.** This type usually carries electrical impulses to the right ventricle of your heart.
- **Dual chamber pacemaker.** This type carries electrical impulses to the right ventricle and the right atrium of your heart to help control the timing of contractions between the two chambers.
- **Biventricular pacemaker.** Biventricular pacing, also called cardiac resynchronization therapy, is for people who have heart failure and heartbeat problems. This type of pacemaker stimulates both of the lower heart chambers (the right and left ventricles) to make the heart beat more efficiently.

## Defibrillators

A defibrillator is a device that provides an electric shock to your heart to allow it to get out of a potentially fatal abnormal heart rhythm, or arrhythmia, — ventricular tachycardia (with no pulse) or ventricular fibrillation — and back to a normal rhythm. Both of these arrhythmias happen in your heart's ventricles or lower chambers.

**Defibrillators for personal use**

A shock from a personal defibrillator can be painless or feel as if someone kicked you in the chest.

People who are at high risk for a life-threatening heart rhythm may have these defibrillator types:

- A **wearable cardioverter defibrillator** you wear like a vest under your clothes. The sensors contact your skin, and the device can send a shock after sensing an abnormal rhythm.

An **implantable cardioverter defibrillator** (ICD). Like an internal watchdog for arrhythmias, an ICD can send the right amount of charge when it's needed. While t's similar to a pacemaker in its ability to keep the heart beating, a defibrillator has the added ability to provide a shock to the heart when a fatal rhythm is detected

## 3.1 Lungs as Purification System: 
Figure 3.12 Representing the oxygen-carbon dioxide exchange in the alveoli and capillary.

### 3.1.1 Lungs as Purifier

The lung purifies air by removing harmful substances and adding oxygen to the bloodstream. The process of purifying air in the lungs can be described as follows:

- Filtration: The nose and mouth serve as a first line of defense against harmful substances in the air, such as dust, dirt, and bacteria. The tiny hairs in the nose, called cilia, and the mucus produced by the respiratory system trap these substances and prevent them from entering the lung

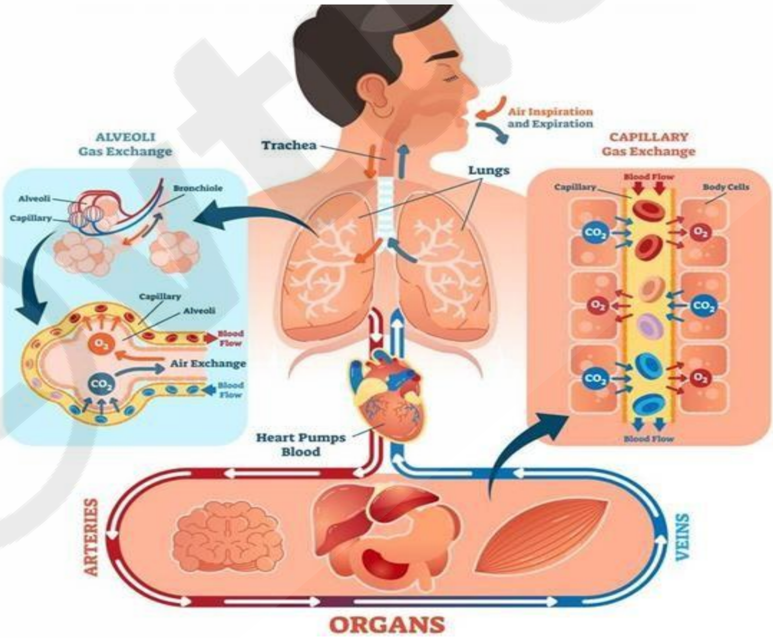


Fig 3.12 Lungs as Purification System

- Moisturization: The air is also humidified as it passes over the moist lining of the respiratory tract, which helps to keep the airways moist and prevent them from drying out.
- Gas Exchange: Once the air reaches the alveoli, the gas exchange process occurs, where oxygen diffuses across the thin alveolar and capillary walls into the bloodstream, and carbon dioxide diffuses in the opposite direction, from the bloodstream into the alveoli to be exhaled. This process ensures that the bloodstream is supplied with fresh, oxygen-rich air, while waste carbon dioxide is removed from the body.

Overall, the lung serves as a vital purification system, filtering out harmful substances, adding oxygen to the bloodstream, and removing waste carbon dioxide. It plays a critical role in maintaining the body's homeostasis and supporting life.

### 3.1.1 Architecture of Lungs as Purification System

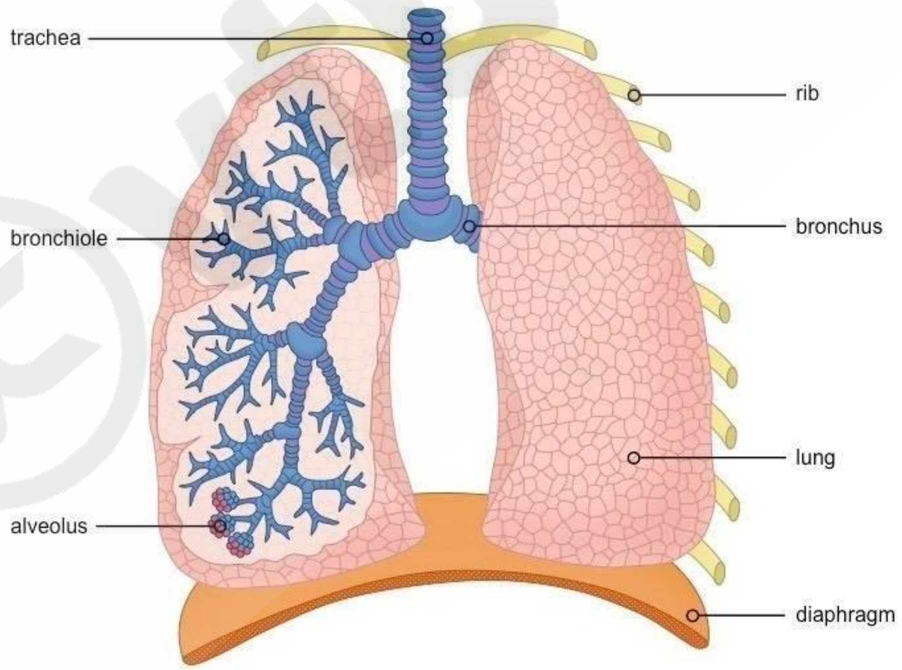


Fig 3.13 Representing structure of lung

The architecture of the lung is designed to maximize surface area for efficient gas exchange. The lung is divided into several parts, including the trachea, bronchi, bronchioles, and alveoli as shown in Fig

3.13.

- Trachea: The trachea is the main airway that leads from the larynx (voice box) to the lungs. It is lined with cilia and mucus-secreting glands that help to filter out harmful substances and trap them in the mucus.
- Bronchi: The trachea branches into two main bronchi, one for each lung. The bronchi are larger airways that continue to branch into smaller airways called bronchioles.
- Bronchioles: The bronchioles are smaller airways that eventually lead to the alveoli. They are surrounded by tiny air sacs called alveoli, which are the sites of gas exchange.
- Alveoli: The alveoli are tiny air sacs that are lined with a network of capillaries. This close proximity of the alveoli and capillaries allows for efficient diffusion of oxygen and carbon dioxide between the air in the alveoli and the bloodstream.

Overall, the architecture of the lung is designed to provide a large surface area for gas exchange, while filtering out harmful substances and humidifying the air. The close proximity of the alveoli and capillaries, along with the moist lining of the respiratory tract, ensures that the air is properly purified and the bloodstream is supplied with fresh, oxygen-rich air.

### 3.1.2 Gas Exchange Mechanism of Lung

The gas exchange mechanism in the lung involves the transfer of oxygen from the air in the alveoli to the bloodstream, and the transfer of carbon dioxide from the bloodstream to the air in the alveoli. This process is known as diffusion and occurs due to differences in partial pressures of oxygen and carbon dioxide.

- Oxygen Diffusion: The partial pressure of oxygen in the air in the alveoli is higher than the partial pressure of oxygen in the bloodstream. This difference creates a gradient that causes oxygen to diffuse from the alveoli into the bloodstream, where it binds to hemoglobin in red blood cells to form oxyhemoglobin.
- Carbon Dioxide Diffusion: The partial pressure of carbon dioxide in the bloodstream is higher than the partial pressure of carbon dioxide in the air in the alveoli. This difference creates a gradient that causes carbon dioxide to diffuse from the bloodstream into the alveoli, where it is exhaled.

### 3.1.3 Spirometry

Spirometry is a diagnostic test that measures the function of the lungs by measuring the amount and flow rate of air that can be exhaled is shown in fig 3.14. The test is commonly used to diagnose lung conditions such as asthma, chronic obstructive pulmonary disease (COPD), and interstitial lung disease.

**Principle:** The principle behind spirometry is to measure the volume of air that can be exhaled from the lungs in a given time period. By measuring the volume of air exhaled, spirometry can provide information about the functioning of the lungs and the ability of the lungs to move air in and out.

**Working:** Spirometry is performed using a spirometer, a device that consists of a mouthpiece, a flow sensor, and a volume sensor. The patient is asked to exhale as much air as possible into the spirometer, and the spirometer measures the volume and flow rate of the exhaled air. The volume of air exhaled is displayed on a graph called a flow-volume loop, which provides information about the lung function.

Fig 3.14 Image of a spirometer

**Interpretation of Results**

The results of spirometry can be used to determine if the lungs are functionning normally and to diagnose lung conditions. For example, a decrease in the volume of air exhaled or a decrease in the flow rate of the exhaled air can indicate a restriction in the airways, which can be a sign of a lung condition such as asthma or COPD. Abnormal Lung Physiology – COPD

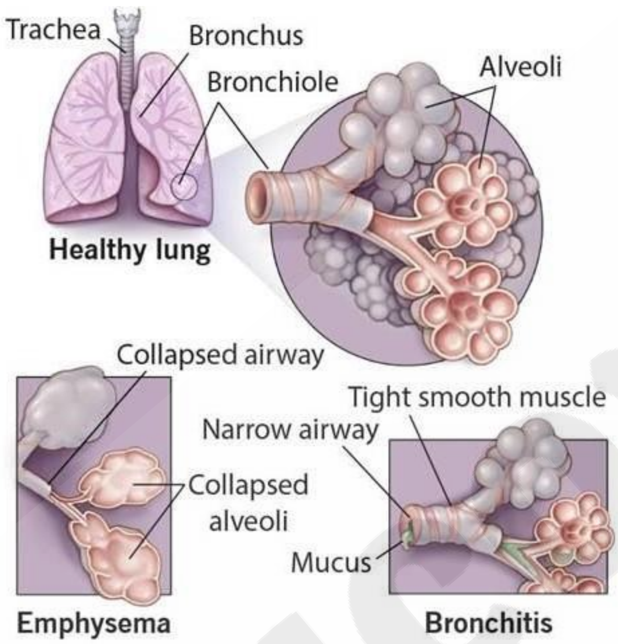


Fig 3.15

Abnormal lung physiology refers to any deviation from the normal functioning of the respiratory system. This can be caused by a variety of factors, including diseases, injuries, or genetic conditions. Some common examples of abnormal lung physiology include:

- Asthma: A chronic inflammatory disease that causes the airways to narrow, making it difficult to breathe.
- Chronic obstructive pulmonary disease (COPD): A progressive lung disease that makes it hard to breathe and can include conditions such as emphysema and chronic bronchitis.
- Pulmonary fibrosis: A disease in which scar tissue builds up in the lungs, making it difficult to breathe and reducing lung function.
- Pneumonia: An infection in the lungs that can cause inflammation and fluid buildup in the air sacs.
- Pulmonary embolism: A blockage in one of the pulmonary arteries, usually by a blood clot, which can cause lung damage and reduce oxygen flow to the body.
- Lung cancer: A type of cancer that originates in the lung and can impair lung function by interfering

with normal air flow and oxygen exchange.

Treatment for abnormal lung physiology depends on the underlying cause and may include medications, lifestyle changes, or surgery.

It's important to seek prompt medical attention if you experience symptoms such as shortness of breath, wheezing, or chest pain, as these can be indicative of a serious lung problem.

### Chronic Obstructive Pulmonary Disease

Chronic Obstructive Pulmonary Disease (COPD) is a group of progressive lung diseases that cause breathing difficulties. It's characterized by persistent airflow limitation that is not fully reversible. The two main forms of COPD are chronic bronchitis and emphysema.

In COPD, the airways and small air sacs (alveoli) in the lungs become damaged or blocked, leading to difficulty in exhaling air. This results in a decrease in lung function, leading to shortness of breath, wheezing, and coughing. Over time, these symptoms can get worse and limit a person's ability to perform everyday activities.

The primary cause of COPD is long-term exposure to irritants such as tobacco smoke, air pollution, and dust. Other risk factors include a history of frequent lung infections, a family history of lung disease, and exposure to second-hand smoke.

There is no cure for COPD, but treatment can help manage the symptoms and slow the progression of the disease. Treatment options include medication, such as bronchodilators and steroids, oxygen therapy, and lung rehabilitation. In severe cases, surgery may also be an option. In addition, quitting smoking and avoiding exposure to irritants is crucial in managing COPD.

## 3.1.4 Ventilators

Ventilators are medical devices used to assist or control breathing in individuals who are unable to breathe adequately on their own. They are commonly used in the treatment of acute respiratory failure, which can occur as a result of a variety of conditions such as pneumonia, severe asthma, and chronic obstructive pulmonary disease (COPD). The ventilator medical device as shown in Fig 3.16

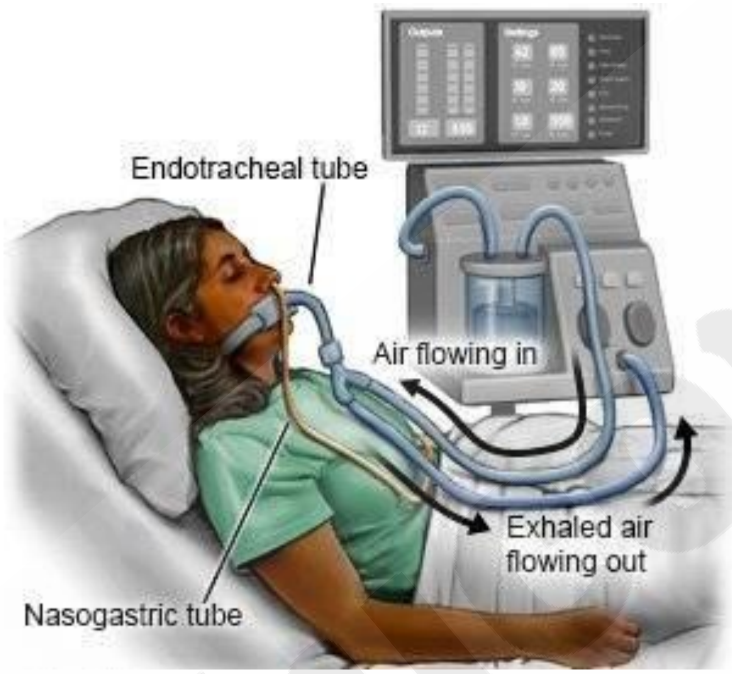


Fig 3. 16 Representing a ventilator machine

There are several different types of ventilators, including volume-controlled ventilators, pressure-controlled ventilators, and bilevel positive airway pressure (BiPAP) devices. The type of ventilator used depends on the patient's individual needs and the type of respiratory failure being treated.

Ventilators work by delivering pressurized air or oxygen into the lungs through a breathing tube or mask. The pressure can be adjusted to match the patient's needs and to help maintain adequate oxygen levels in the blood.

While ventilators can be lifesaving for individuals with acute respiratory failure, they also come with potential risks and complications. For example, prolonged use of a ventilator can increase the risk of ventilator-associated pneumonia, and patients may experience discomfort or pain from the breathing tube. The use of ventilators is carefully monitored and managed by healthcare professionals to ensure that the patient receives the appropriate level of support while minimizing potential risks and complications.

### 3.1.5 Heart-Lung Machine

A heart-lung machine, also known as a cardiopulmonary bypass machine, is a device used in cardiovascular surgery to temporarily take over the functions of the heart and lungs. The heart-lung machine is used during open-heart surgery, such as coronary artery bypass graft (CABG) surgery and valve replacement surgery, to support the patient's circulatory and respiratory functions while the heart is stopped.

The heart-lung machine works by circulating blood outside of the body through a series of tubes and pumps. Blood is taken from the body, oxygenated, and then returned to the body. This allows the heart to be stopped during the surgery without causing any harm to the patient.

The use of a heart-lung machine during surgery carries some risks, including the potential for blood clots, bleeding, and infections. Additionally, there may be some long-term effects on the body, such as cognitive decline, that are not yet fully understood. However, the use of a heart-lung machine has revolutionized the field of cardiovascular surgery, allowing for more complex procedures to be performed and greatly improving patient outcomes.

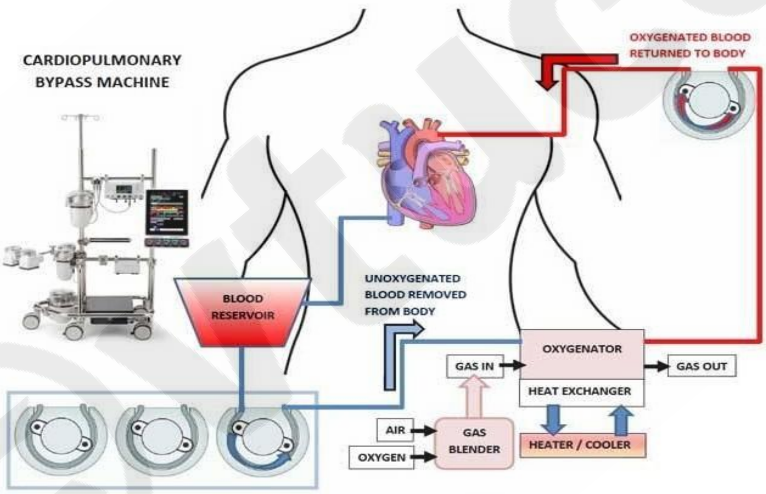


Fig 3.17 Representing a heart-lung machine

### 3.1.6 Artificial Lungs

Artificial lungs are devices designed to mimic the function of the natural respiratory system. They are used to support patients with acute respiratory distress syndrome (ARDS) or acute lung injury (ALI) and to help the patient's own lungs recover and heal.

### Types

There are two main types of artificial lungs: membrane oxygenators and extracorporeal lung assist devices.

Membrane Oxygenators: These are devices that use a semipermeable membrane to transfer oxygen and carbon dioxide between the blood and the air. The blood is pumped through the membrane, where it comes into contact with air, allowing for the exchange of gases as shown in Fig 3.18.

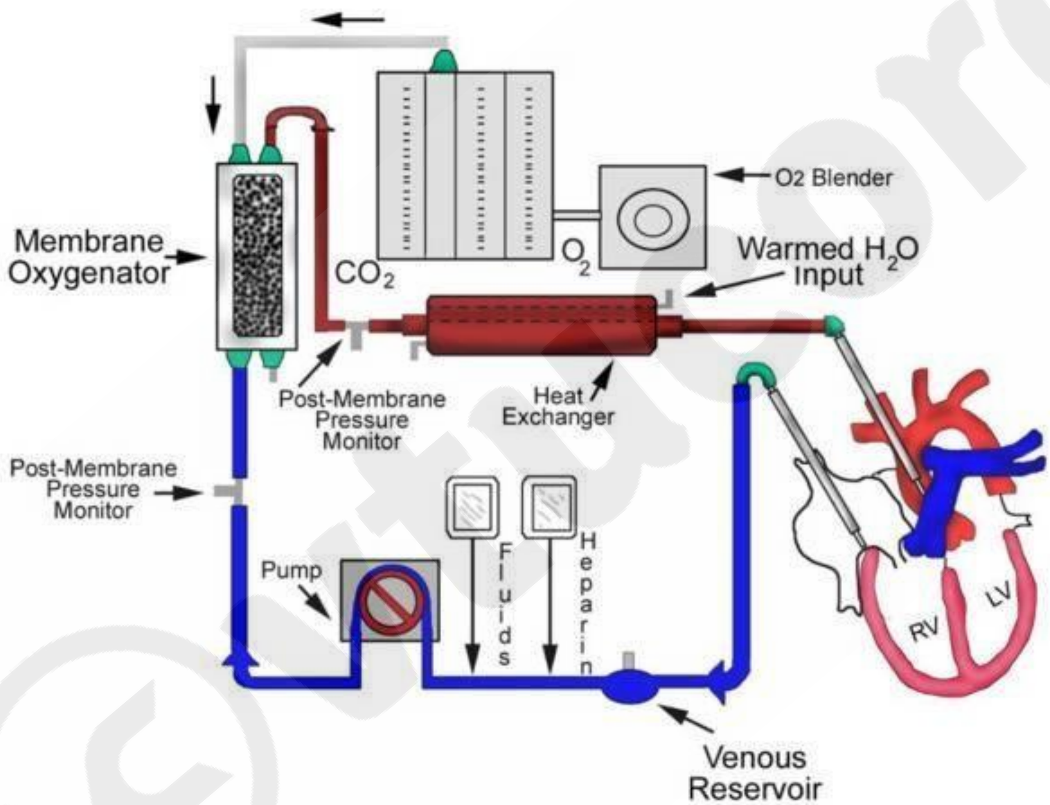


Fig 3.18 Representing a membrane oxygenator

Extracorporeal Lung Assist Devices: These devices work by removing carbon dioxide from the blood and adding oxygen, allowing the patient's natural lungs to rest and heal. One example of an extracorporeal lung assist device is the extracorporeal membrane oxygenation (ECMO) machine, which is used to treat patients with severe respiratory failure. ECMO works by removing carbon dioxide from the blood and adding oxygen, and it can be used as a bridge to recovery or as a bridge to lung transplantation.

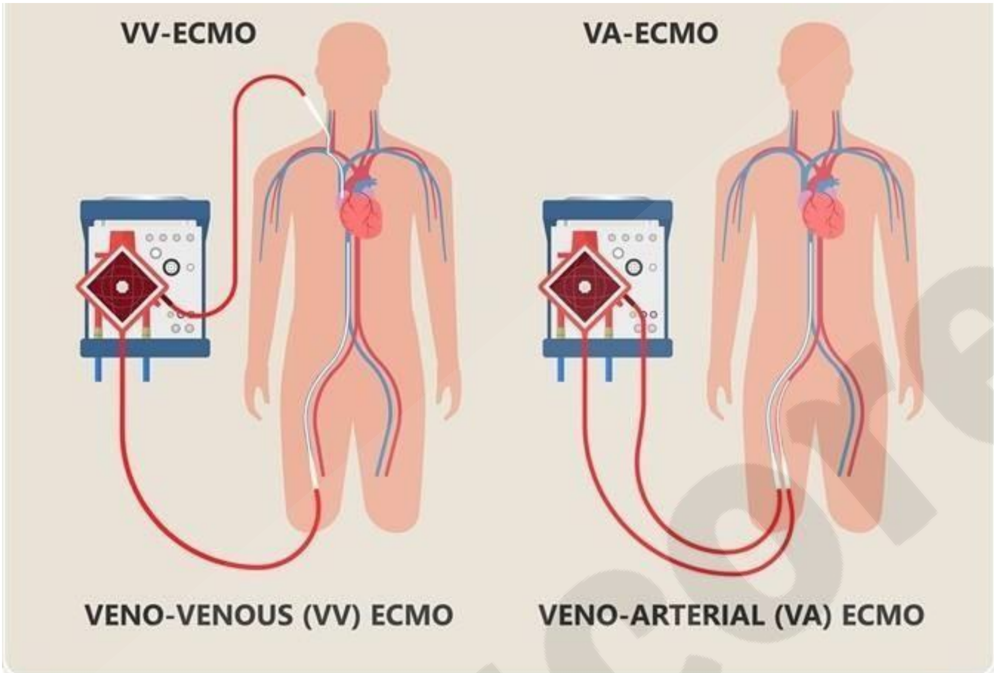


Fig 3.19 Representing veno-venous and veno-arterial extracorporeal membrane oxygenation

## 3.2 Kidney as a Filtration System:

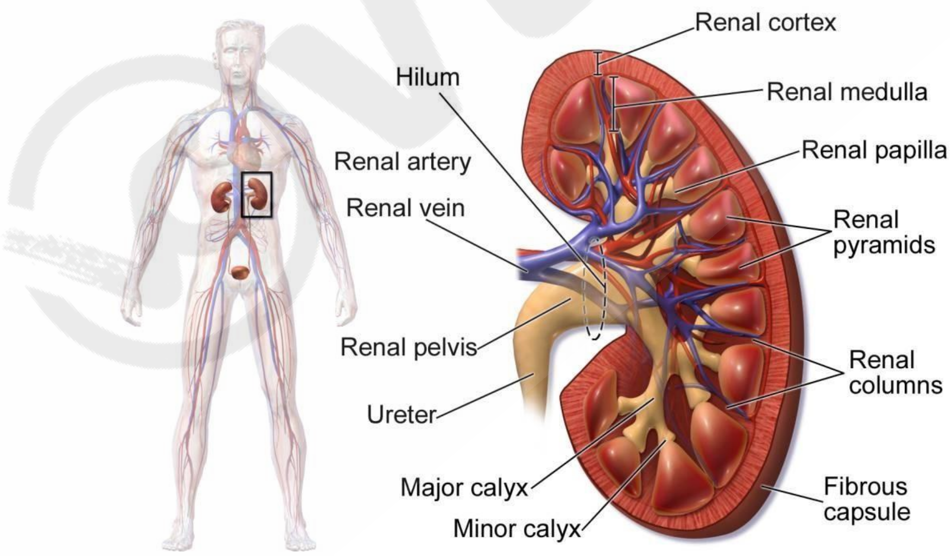


Fig 3.20 Anatomy of kidney

The kidney is a complex organ that acts as a filtration system for the body as shown in Fig 3.20. It removes waste and excess fluid from the bloodstream and maintains a delicate balance of electrolytes, hormones, and other substances that are critical for the body's normal functioning. The kidney also plays an important role in regulating blood pressure by secreting the hormone renin, which helps control the balance of fluid and electrolytes in the body. It also regulates red blood cell production and the levels of various minerals in the blood, such as calcium and phosphorus.

Without the kidney, waste and excess fluid would accumulate in the body, leading to serious health problems.

### 3.2.1 Architecture of Kidney

The kidney is composed of functional units called nephrons, which are the basic structural and functional units of the kidney. Each kidney contains approximately one million nephrons, and each nephron performs the functions of filtration, reabsorption, and secretion.

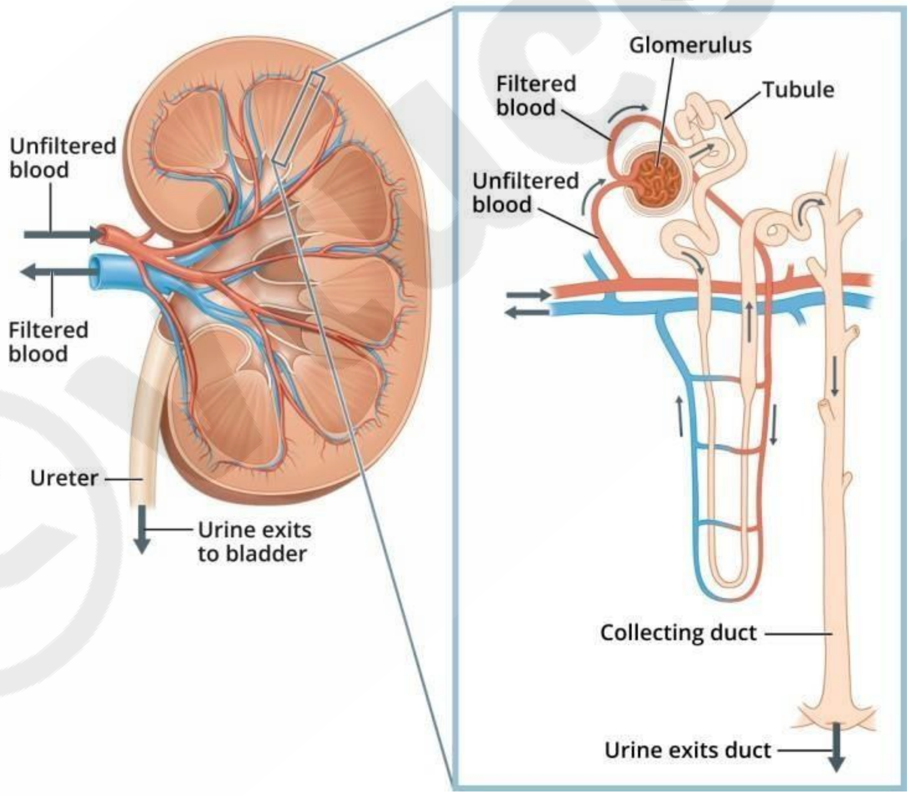


Fig 3.21 Representing kidney and nephron

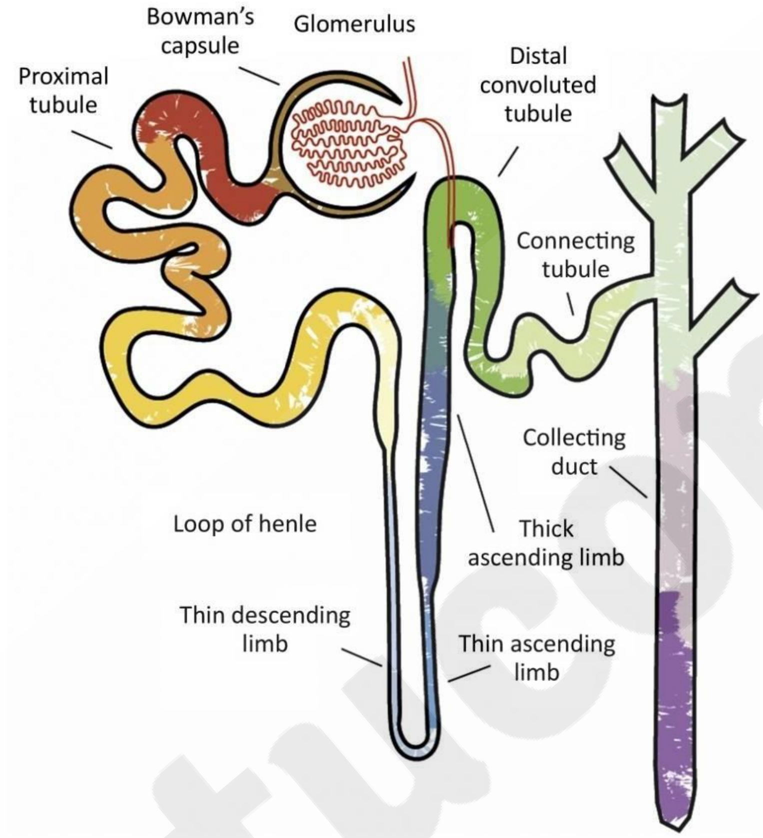


Fig 3.22: Representing the parts of nephron

the nephron is comprised of several key structures:

- Bowman's capsule: This is a cup-shaped structure that surrounds the glomerulus and filters waste and excess fluid from the bloodstream into the renal tubule.
- Glomerulus: A network of tiny blood vessels within the Bowman's capsule that filters waste and excess fluid from the bloodstream.
- Proximal convoluted tubule: A segment of the renal tubule that reabsorbs important substances, such as glucose, amino acids, and electrolytes, back into the bloodstream.
- Loop of Henle: A U-shaped segment of the renal tubule that is critical for the reabsorption of ions and water.
- Distal convoluted tubule: A segment of the renal tubule that regulates the levels of

electrolytes and other important substances in the bloodstream.

- Collecting duct: A series of ducts that collect the filtrate from the renal tubules and transport it to the renal pelvis, where it drains into the ureter and eventually into the bladder.

The nephrons are surrounded by a network of blood vessels, including the afferent arteriole and the efferent arteriole, which bring blood into and out of the glomerulus, respectively. The filtrate produced by the nephron passes through the renal tubules, where it is modified by reabsorption and secretion, before being eliminated from the body as urine.

### 3.2.2 Mechanism of Filtration – Urine Formation

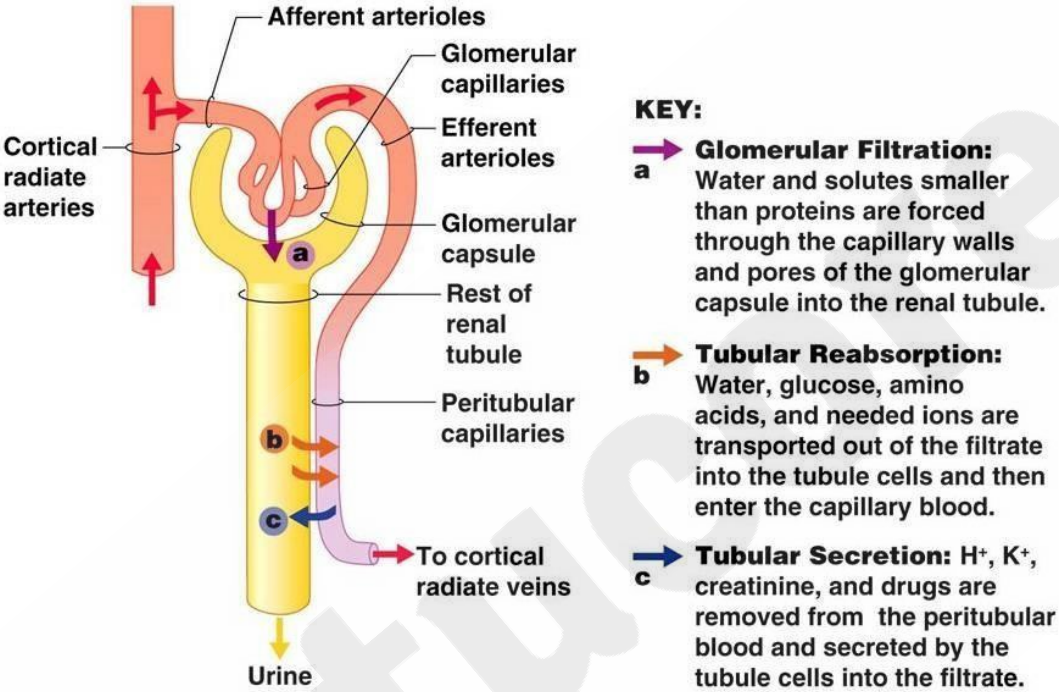


Fig 3.23 Schematic of mechanism of filtration in human kidney

The mechanism of filtration in the kidneys is a complex process that involves multiple steps to remove waste and excess fluids from the bloodstream. The following is a summary of the steps involved in the filtration process:

- Blood enters the kidney through the renal arteries and flows into tiny filtering units called glomeruli.
- At the glomerulus, the pressure in the blood vessels causes a portion of the plasma and dissolved substances to filter out and enter a structure called Bowman's capsule.
- In Bowman's capsule, the filtrate is then transferred into the renal tubules, which are the main filtering units of the kidneys.
- In the renal tubules, the filtrate passes through a series of specialized cells, such as proximal tubular cells and distal tubular cells, which reabsorb important substances such as glucose, amino acids, and electrolytes back into the bloodstream.

- At the same time, the renal tubules secrete waste products, such as urea and creatinine, back into the filtrate.
- Finally, the filtered fluid, now known as urine, is transported through the renal pelvis and ureters to the bladder, where it is eventually eliminated from the body.

This process of filtration, reabsorption, and secretion helps to maintain the proper balance of fluids and electrolytes in the body, as well as to remove waste and excess substances.

### 3.2.3 Chronic Kidney Disease (CKD)

CKD stands for Chronic Kidney Disease. It is a long-term condition in which the kidneys gradually become less able to function properly. It can be caused by a variety of factors, including diabetes, high blood pressure, and other health problems that damage the kidneys.

Symptoms of CKD include fatigue, swelling in the legs and feet, trouble sleeping, and difficulty concentrating. As the disease progresses, it can lead to more serious complications, such as anemia, nerve damage, and an increased risk of heart disease and stroke.

Treatment for CKD may include lifestyle changes, such as eating a healthy diet and exercising regularly, as well as medications to manage symptoms and underlying health conditions. In severe cases, kidney transplant or dialysis may be necessary.

It is important for individuals with risk factors for CKD to get regular check-ups and to talk to their doctor about how to best manage their condition.

### 3.2.4 Dialysis Systems

Dialysis is a medical treatment that helps to filter waste and excess fluids from the blood when the kidneys are unable to function properly. There are two main types of dialysis systems: hemodialysis and peritoneal dialysis.

Hemodialysis is a procedure that uses a machine to clean the blood. During hemodialysis, blood is removed from the body, passed through a dialysis machine that filters out waste and excess fluids, and then returned to the body. Hemodialysis typically takes place in a hospital or dialysis center, and is typically performed three times a week for three to four hours at a time.

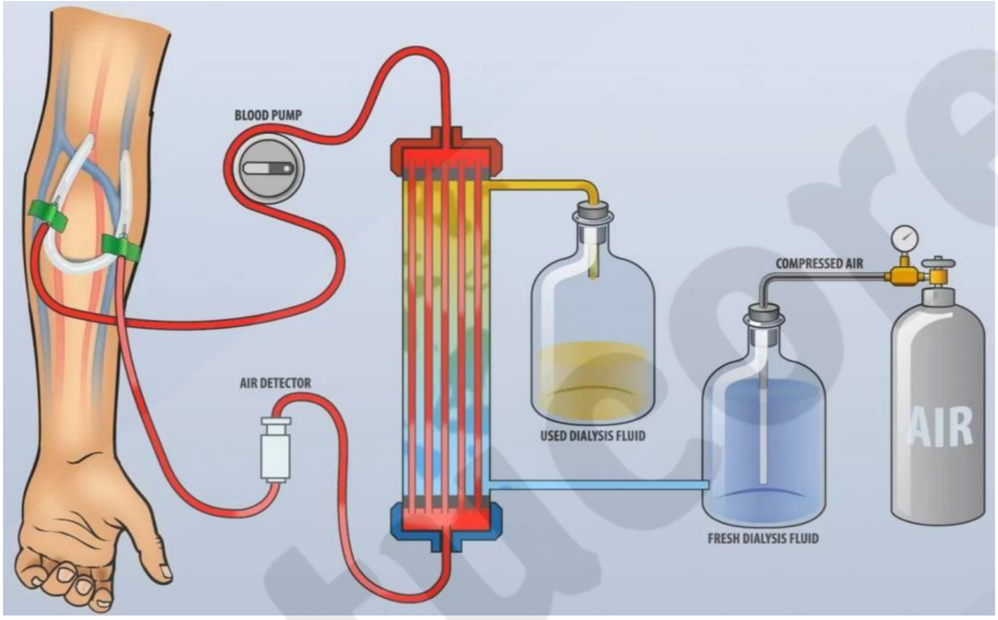


Fig 3.24 Representing a Hemodialysis

Peritoneal dialysis is a type of dialysis that uses the lining of the abdomen, called the peritoneum, to filter waste and excess fluids from the blood. A sterile solution is introduced intothe abdomen, where it absorbs waste and excess fluids, and is then drained and replaced with fresh solution. Peritoneal dialysis can be performed at home and allows for more flexibility in scheduling.

Both hemodialysis and peritoneal dialysis can effectively treat the symptoms of kidney failure, but each has its own advantages and disadvantages. The choice of dialysis system depends on various factors such as the individual's overall health, lifestyle, and personal preferences.

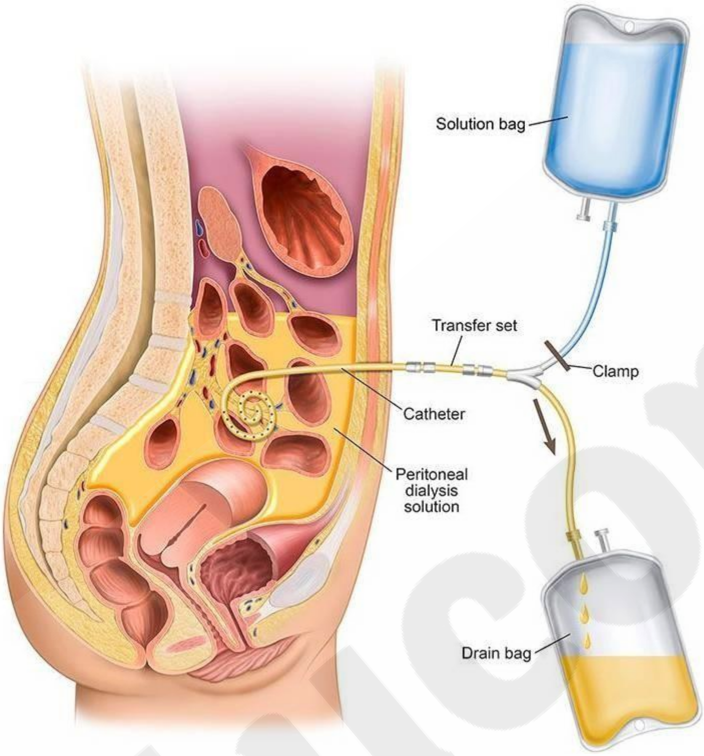

Fig 3.25 Representing a Peritoneal dialysis

### 3.2.5 Artificial Kidney

While much progress has been made in developing an artificial kidney, it is still in the experimental stage and is not yet widely available. Further research and development is needed to improve the efficiency and safety of artificial kidney devices, and to ensure that they can be widely adopted as a treatment for chronic kidney disease.

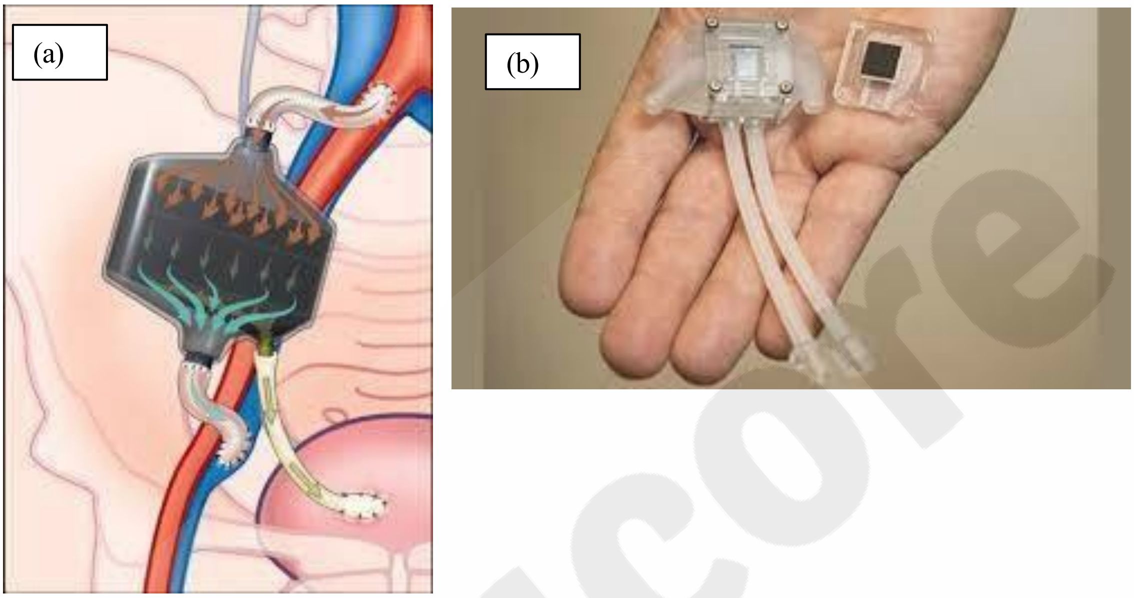


Fig 3.26 a) Schematic representation b) a prototype of artificial kidney

An artificial kidney is a device that is being developed to mimic the functions of the human kidney. The goal of an artificial kidney is to provide a more effective and efficient means of treating patients with chronic kidney disease, who currently rely on dialysis or kidney transplantation.

There are currently two main approaches to developing an artificial kidney: a biological approach and a technological approach.

The biological approach involves using living cells, such as kidney cells or stem cells, to create a functional, implantable artificial kidney.

The technological approach involves using synthetic materials, such as silicon or polymer, to create a dialysis device that can filter the blood and remove waste and excess fluids.

It's important to note that while the development of an artificial kidney holds great promise, it is not a cure for chronic kidney disease and patients with kidney failure will still need dialysis or kidney transplantation in the meantime.